Nearest-Neighbor LASSO Logistic Regression for Gradient

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Background

- Estimating the gradient of a regression function is an appealing problem in nonparametric statistics and machine learning.
- Gradient estimation plays a key role in uncovering structural properties of high-dimensional data

Binary Classification Problem

Given predictors $X \in \mathbb{R}^p$ and binary outcome $Y \in \{0, 1\}$,
$\Rightarrow$ In general, we study conditional probability.

$$\pi(x) = P(Y = 1 \mid X = x)$$

Define the logit transformation:

$$\ell(x) = \log\left(\frac{\pi(x)}{1 - \pi(x)}\right).$$

Challenges with gradient estimation in Local Linear Estimation

We are interested in the gradient estimation of π as $\Delta\ell(x)$.
$\Rightarrow$ **Local Linear Estimation** (Fan and Gijbels, 1996)

For observations near x,

$$\ell(X_i) \approx a + b^T(X_i - x), \text{where} \qquad a = \ell(x), \qquad b = \nabla\ell(x).$$

The parameters are obtained by minimizing.

$$(\widehat{a}(x), \widehat{b}(x)) = \arg\min_{a,b} \sum_{i=1}^{n} \left[Y_i - a - b^\top(X_i - x)\right]^2 K_h(X_i - x),$$

where $K_h(\cdot)$ is a kernel weight function.

- Local first-order approximation
- Kernel-based weighting
- $\widehat{b}(x)$ estimates the gradient
- Suffers in high dimensions due to sparse local neighborhoods

Proposed Method: Penalized Local Logistic Estimation

We introduce a nearest-neighbor, penalized, local logistic estimator of $\pi(x)$ and its gradient as

$$(\widehat{a}_n(x), \widehat{b}_n(x)) = \arg\max_{a,b} \Bigg\{ \sum_{i \in N_k(x)} Y_i(a + b^T(X_i - x)) - \log(1 + \exp(a + b^T(X_i - x))) - \lambda \|b\|_1 \Bigg\}.$$

Main Contributions

- Logistic local likelihood for classification
- k-nearest neighbor localization
- ℓ_1-penalization induces sparsity
- Stable and interpretable in high dimensions

Outer Product of Gradients for Dimension Reduction

$$\pi(x) = P(Y = 1 \mid X = x) = g(\beta^\top x), \qquad \beta \in \mathbb{R}^{p\times d},\ d \ll p.$$

Since

$$\nabla \ell(x) \in \mathrm{span}(\beta), \qquad \ell(x) = (\pi(x)),$$

the **central subspace** can be recovered from local gradients.

$\Rightarrow$**Outer Product of Gradients Matrix**

$$\widehat{M} = \frac{1}{m} \sum_{i=1}^{m} \widehat{b}_n(X_i^*) \widehat{b}_n(X_i^*)^\top.$$

The leading eigenvectors of $\widehat{M}$ estimate the central subspace.
The projection matrix is defined

$$\widehat{P}_{\widehat{\beta}} = \widehat{P}_{\widehat{\beta},d} = \sum_{k=1}^{d} \widehat{\beta}_k \widehat{\beta}_k^T$$

Theoretical contributions vs. existing methods

Property	**Our Method**	**wOPG (Kang & Shin 2022)**
Gradient Estimation	Nearest-neighbor logistic	RKHS large-margin method
Variable Selection	Yes (ℓ_1-penalized)	No
Weak Convergence Theory	Yes	Not established
Binary Classification SDR	Yes	Yes
Local Estimation Method	Nearest neighbors	Kernel / RKHS

Algorithm: Estimation of the dimension reduction matrix

1: **Input:** $\{(X_i, Y_i)\}_{i=1}^n \in \mathbb{R}^p \times \{0,1\}$, $\lambda > 0$, $k \in \{1,\dots,n\}$, $m \in \{1,\dots,n\}$

2: **Output:** Dimension reduction matrix $\widehat{M}$

3: Draw uniformly a subset $\mathcal{X}_m$ of m observations from $\mathcal{X} = (X_1,\dots,X_n)$

4: **for** each $x \in \mathcal{X}_m$ **do**

5: Compute $N_k(x)$, the indices of the k nearest neighbors of x

6: Estimate $\widehat{a}_n(x)$ and $\widehat{b}_n(x)$ using the penalized local logistic estimator

7: **end for**

8: Compute

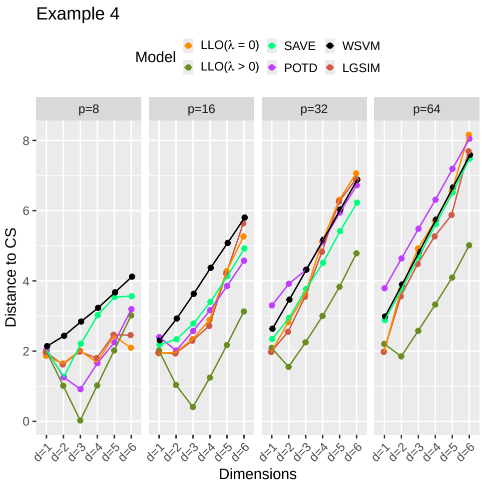
$$\widehat{M} = \frac{1}{m} \sum_{x \in \mathcal{X}_m} \widehat{b}_n(x) \widehat{b}_n(x)^\top$$

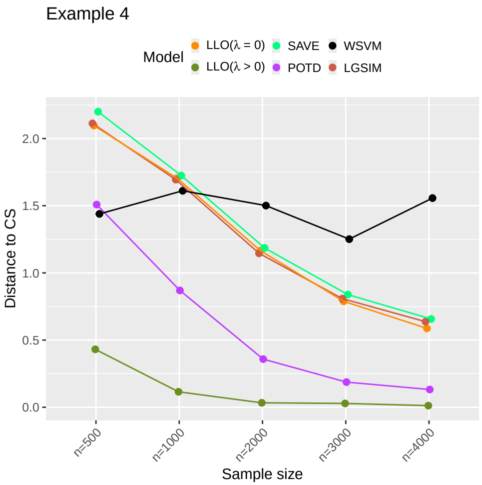
9: Return $\widehat{M}$

Synthetic data application

Data Generation (with $p = 8, 16, 32, 64$ covariates):

- Example 4: $Y = \text{sign}(\log(X_1^2)(X_2^2 + X_3) + 0.2\epsilon)$, where $\epsilon \sim \mathcal{N}(0, 1)$.





Summary

What we contributed:

1. **Theory:**
   - Central limit theorem for nearest-neighbor empirical processes
   - Weak convergence of gradient estimators at **minimax optimal rate**
   - Extension to multi-class GLMs
2. **Methods:**
   - Nearest-neighbor LASSO logistic regression for gradient estimation
   - Dimension selection via cross-validation
3. **Practice:**
   - Consistent improvements on synthetic and real data
   - Robust performance in high dimensions
   - Actionable algorithm for practitioners

References

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Thank You!

Questions?